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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B. Tech (CBCS) Degree Examinations

May 2025

VT3102-1 - MICROFABRICATION AND MEMS

(For VL)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What is one of the main advantages of integrated microsystems over traditional systems?
A) Larger size
B) Increased complexity
C) Miniaturization and reduced power consumption
D) Higher manufacturing costs
2. In the Czochralski crystal growth process, the materials are heated up to
A) 950°C
B) 1050°C
C) 1420°C
D) 1240 °C
3. Which of the following is the primary goal of the PVD process?
A) To create thin, smooth films on substrates
B) To perform surface cleaning
C) To deposit a layer of photoresist
D) To etch patterns into a surface
4. Which material is commonly used as the mask in photolithography?
A) Quartz
B) Silicon
C) Chromium on glass
D) Aluminum oxide
5. If lateral etch rate=0, the degree of anisotropy will be equal to _____
A) 0
B) 1
C) 2
D) 4
6. Which of the following is required to initiate anodic bonding?
A) High temperature and electric field
B) Low pressure and chemical etching
C) Low temperature and vacuum conditions
D) High pressure and UV radiation
7. In the equation for Fick's First Law, what does "D" represent?
A) Concentration gradient
B) Diffusion coefficient
C) Diffusion flux
D) Diffusion time
8. Which of the following is an advantage of CVD over PVD?
A) Lower deposition rates
B) Ability to deposit films on complex surfaces
C) Use of lower temperatures
D) Ability to use more diverse materials
9. Which of the following is a key advantage of MEMS magnetic micro relays?
A) High power consumption
B) Large physical size
C) High voltage tolerance
D) Fast switching time and low power requirement
10. What MEMS device commonly uses a Wheatstone Bridge for its operation?
A) Accelerometer
B) Capacitive sensor
C) Optical sensor
D) Pressure sensor

11. What is the primary purpose of a MEMS gyroscope?
 A) To measure temperature B) To measure electrical resistance
 C) To measure orientation or angular velocity D) To measure light intensity
12. In Micromirror Array, Gold is used for
 A) Better mirror-reflectivity B) Light exposure
 C) Mechanical stress D) Thermal changes
13. Which component is commonly used in MEMS signal conditioning to amplify the sensor's output?
 A) Transformer B) Op-Amp
 C) Capacitor D) Inductor
14. In an instrumentation amplifier, the output voltage is directly proportional to
 A) The difference between the two input voltages B) The average of the input voltages
 C) The sum of the two input voltages D) The product of the input voltages
15. In an N-bit ADC, how many distinct values can the output signal take?
 A) N values B) 2^N values
 C) N^2 values D) $2^N - 1$ values
16. Instrumentation amplifiers are widely used in:
 A) Power supply design B) Lighting control circuits
 C) Audio processing systems D) Medical instrumentation
17. Piezoelectric materials generate an electrical charge when subjected to:
 A) Magnetic fields B) Light exposure
 C) Mechanical stress D) Thermal changes
18. Which of the following is the principle behind a MEMS accelerometer?
 A) Hall effect B) Thermal sensing
 C) Capacitive sensing D) Piezoelectric effect
19. Which type of RF MEMS device is used for switching high-frequency signals?
 A) MEMS switches B) MEMS pressure sensors
 C) MEMS accelerometers D) MEMS microphones
20. What type of actuation is commonly used in RF MEMS switches?
 A) Magnetic actuation B) Electrostatic actuation
 C) Thermal actuation D) Piezoelectric actuation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain briefly the general classification of integrated microsystem.	4	L*2	1	1
	b) The equation of a plane is given by $7x + 3y = 2$. Obtain the Miller indices for a plane that is normal to this plane.	4	L3	2	1
	c) With relevant sketches, explain the different CVD techniques used for thin film deposition.	8	L2	2	1
2.	a) Briefly explain the significance of microfabrication	4	L2	1	1
	b) Briefly explain wafer bonding and the different types of wafer bonding	4	L2	2	1
	c) Explain LIGA process in detail	8	L2	2	1
3.	a) Explain the purpose of various components used in a smart system.	4	L2	1	1
	b) With the help of a neat sketch explain the ion implantation process in detail. Also mention its advantages and disadvantages.	8	L2	2	1
	c) Give a comparison on dry etching and wet etching	4	L2	2	1

Unit – II

4. a) With neat sketch, explain the principle of operation of a piezoelectric inkjet actuator.
 b) Derive an expression for the gain and input resistance of a difference amplifier.
5. a) With relevant diagrams, explain the principle of operation of a gas sensor. Also mention the typical application of a gas sensor.
 b) In the instrumentation amplifier in Fig. 5(b) the resistance values are $R_1 = 50 \text{ k}\Omega$, $R_5 = 20 \text{ k}\Omega$ and $R_4 = 10 \text{ k}\Omega$. The resistance R_2 is a series combination of a fixed resistance R_f and a potentiometer having maximum value R_p . Determine R_f and the range of R_p so that the differential gain A_d can be varied from 6 to 202.

8 L2 3 1

8 L2 4 1

8 L2 3 1

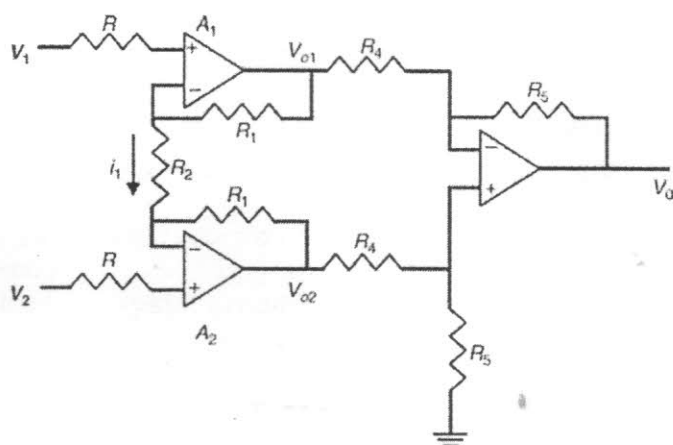


Fig. 5(b)

8 L2 4 1

6. a) With a neat sketch, explain the principle of operation of a fiber optic sensor. Mention the advantages of fiber optic sensors.
 b) Derive an expression for the output voltage of an instrumentation amplifier

8 L2 3 1

8 L2 4 1

Unit – III

7. a) Explain in detail the operation of a RF switch. Also give the figure of merit of RF MEMS
 b) Explain with relevant figures, the different sensing schemes used in inertial sensors.
8. a) Explain the principle of operation of a piezoresistive pressure sensor. Also mention the advantages and typical application of a piezoresistive pressure sensor
 b) Determine (a) the burst pressure and (b) the maximum pressure up to which a pressure sensor works in the linear range of operation in the following three cases for square silicon diaphragms with thickness h and diaphragm side length $2a$: (i) $h = 5 \text{ }\mu\text{m}$, $a = 250 \text{ }\mu\text{m}$
 (ii) $h = 10 \text{ }\mu\text{m}$, $a = 250 \text{ }\mu\text{m}$.

8 L2 5 1

8 L2 5 1

10 L2 5 1

6 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
